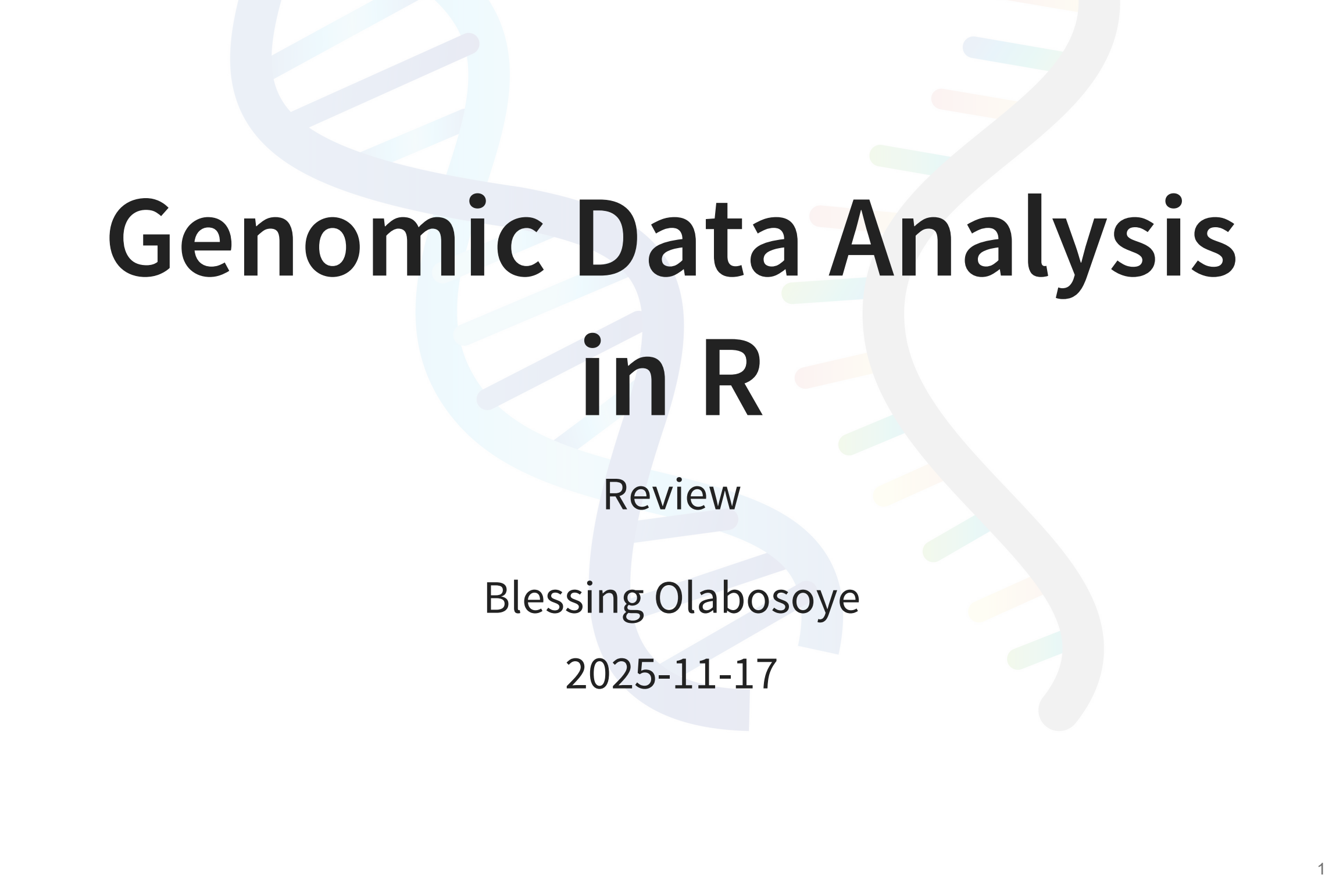




Genomic Data Analysis in R

Review

Blessing Olabosoye

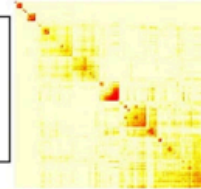
2025-11-17

WGCNA

Construct a gene co-expression network

Rationale: make use of interaction patterns among genes

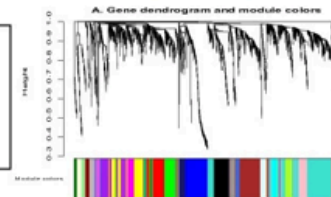
Tools: correlation as a measure of co-expression



Identify modules

Rationale: module (pathway) based analysis

Tools: hierarchical clustering, Dynamic Tree Cut

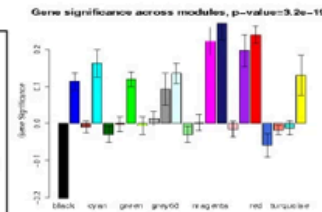


Relate modules to external information

Array Information: clinical data, SNPs, proteomics

Gene Information: ontology, functional enrichment

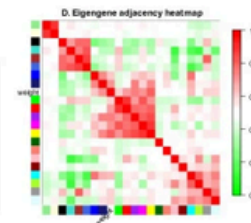
Rationale: find biologically interesting modules



Study module relationships

Rationale: biological data reduction, systems-level view

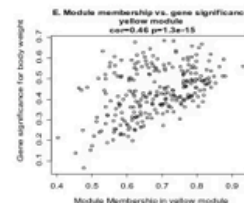
Tools: Eigengene Networks



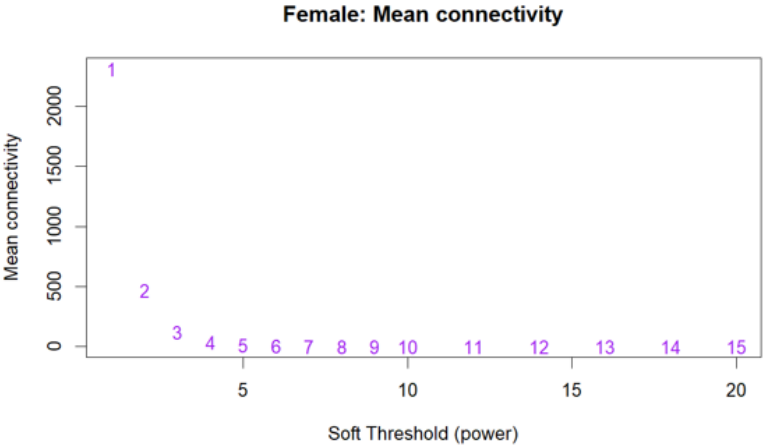
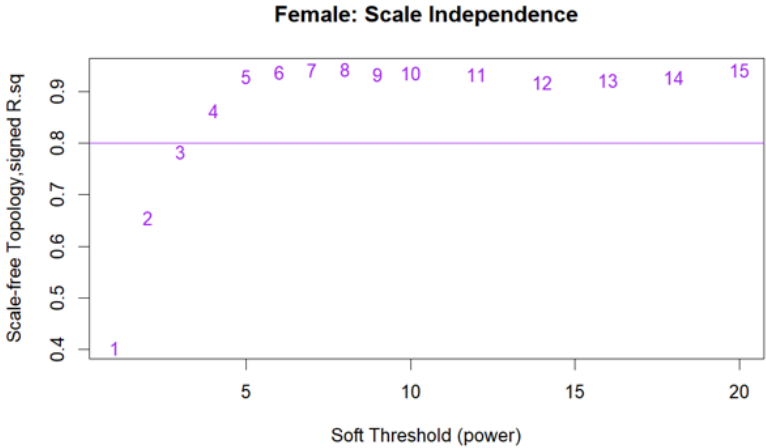
Find the key drivers in *interesting* modules

Rationale: experimental validation, biomarkers

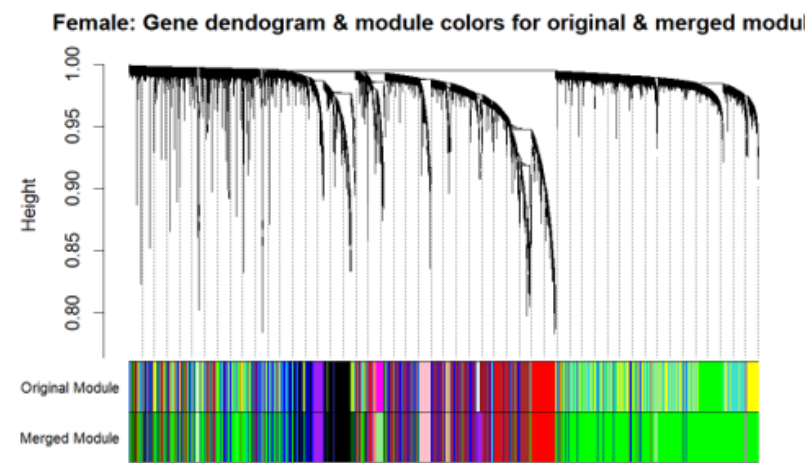
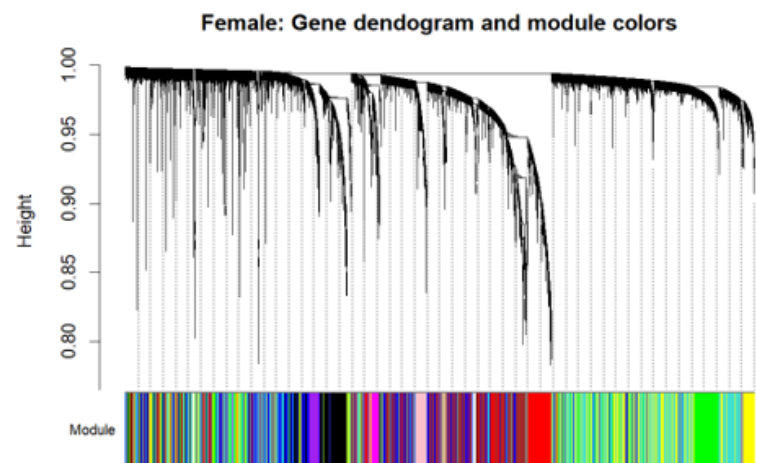
Tools: intramodular connectivity, causality testing



Power selection

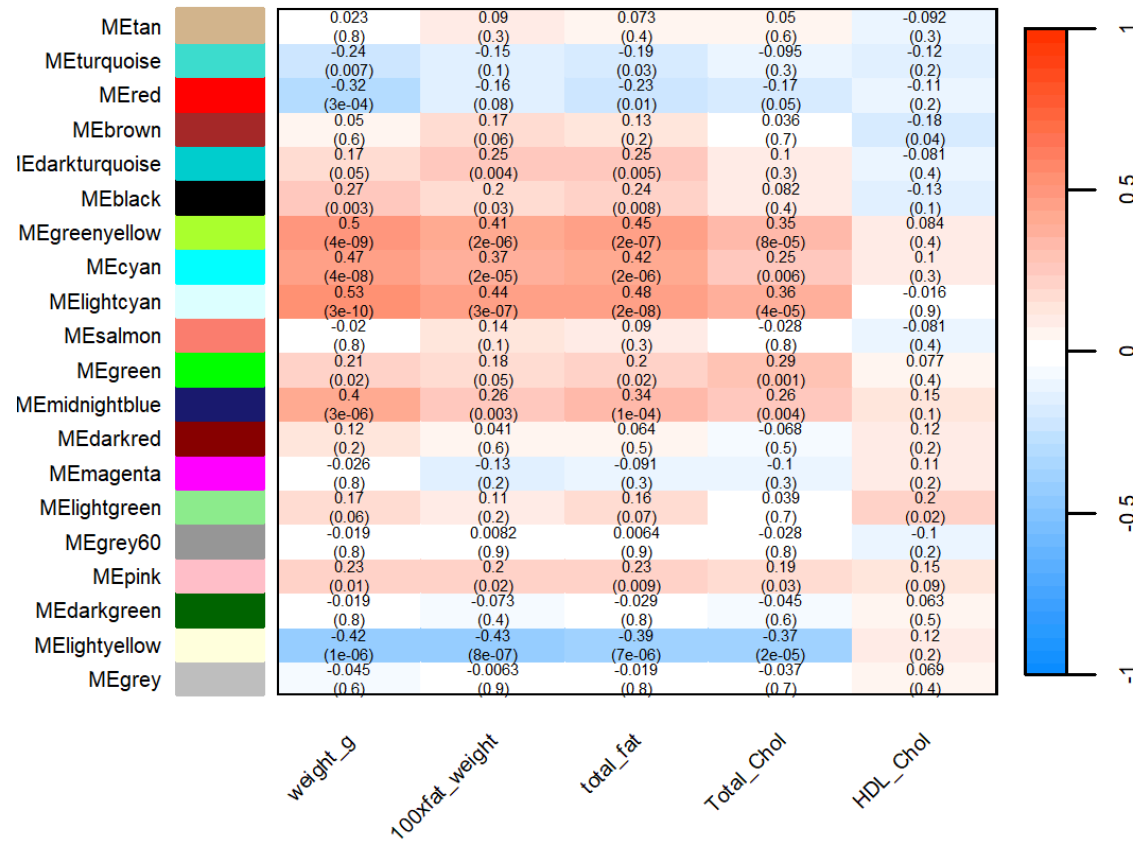


Identify gene modules

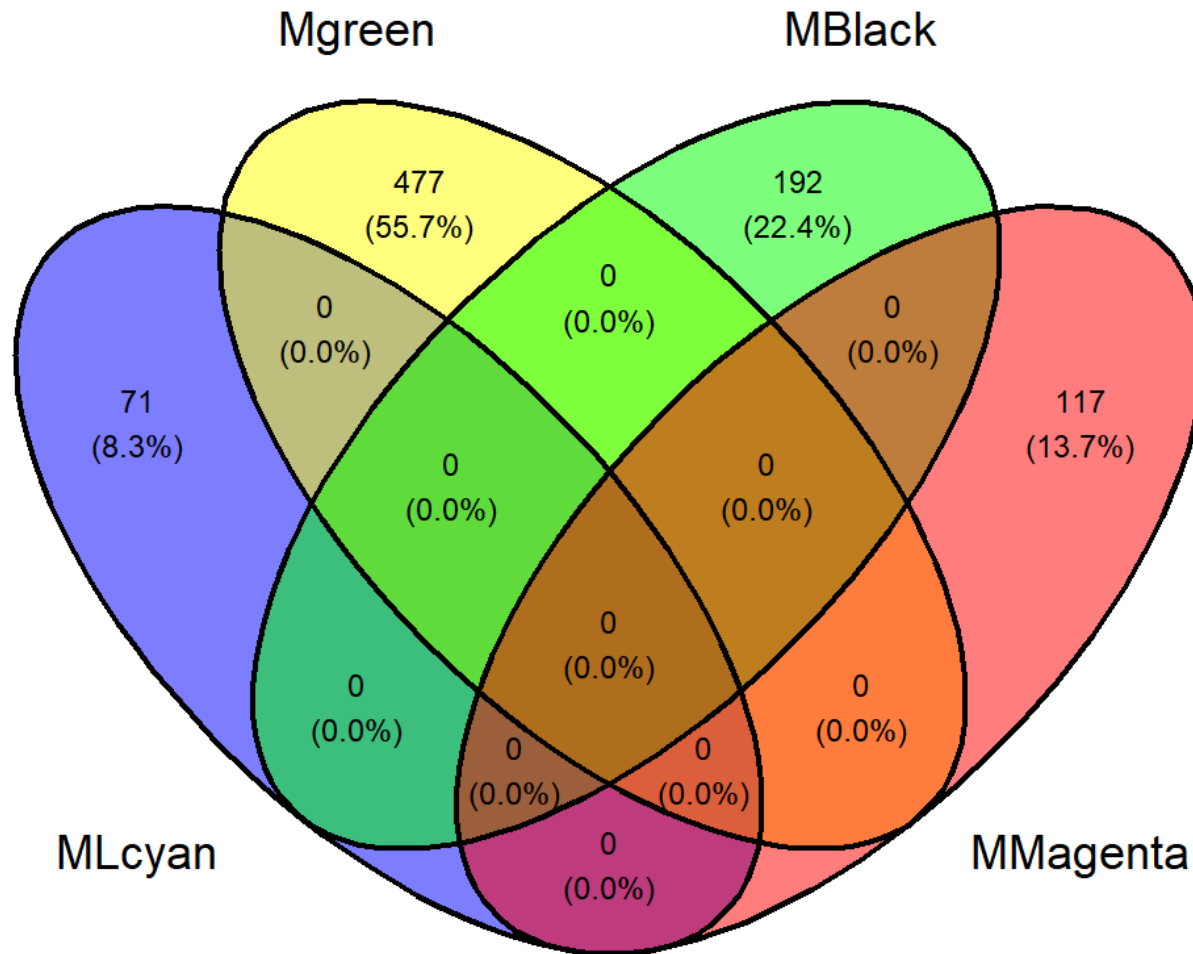


Relate modules to trait

Female: Module-trait relationships



Visualize number of genes in top 4 module



Polygenic score

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{g} + \mathbf{e}$$

where, $\mathbf{y} \in \mathbb{R}^n$ is the phenotype of interest denoted as response variable

- Fixed effects
 - $\mathbf{X}$ is the $n \times p$ incidence/design matrix containing an intercept
 - $\boldsymbol{\beta} \in \mathbb{R}^p$ is the vector of fixed effect(s) e.g. sex and batch on the phenotype
- Random effects
 - $\mathbf{Z}$ is the $n \times m$ incidence/design matrix linking
 - $\mathbf{g} \in \mathbb{R}^m$ the vector of random effect(s) e.g. allele substitution effect on the phenotype with a normal distribution $\mathbf{g} \sim \mathcal{N}(0, \sigma^2 \mathbf{g}\mathbf{I})$

- Random effects (cont'd)
 - $\mathbf{e} \in \mathbb{R}^n$ is the vector of residual variable with a normal distribution
 $\mathbf{e} \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$

Note that β , $\mathbf{g}$, and $\mathbf{e}$ are unknown and will be estimated.

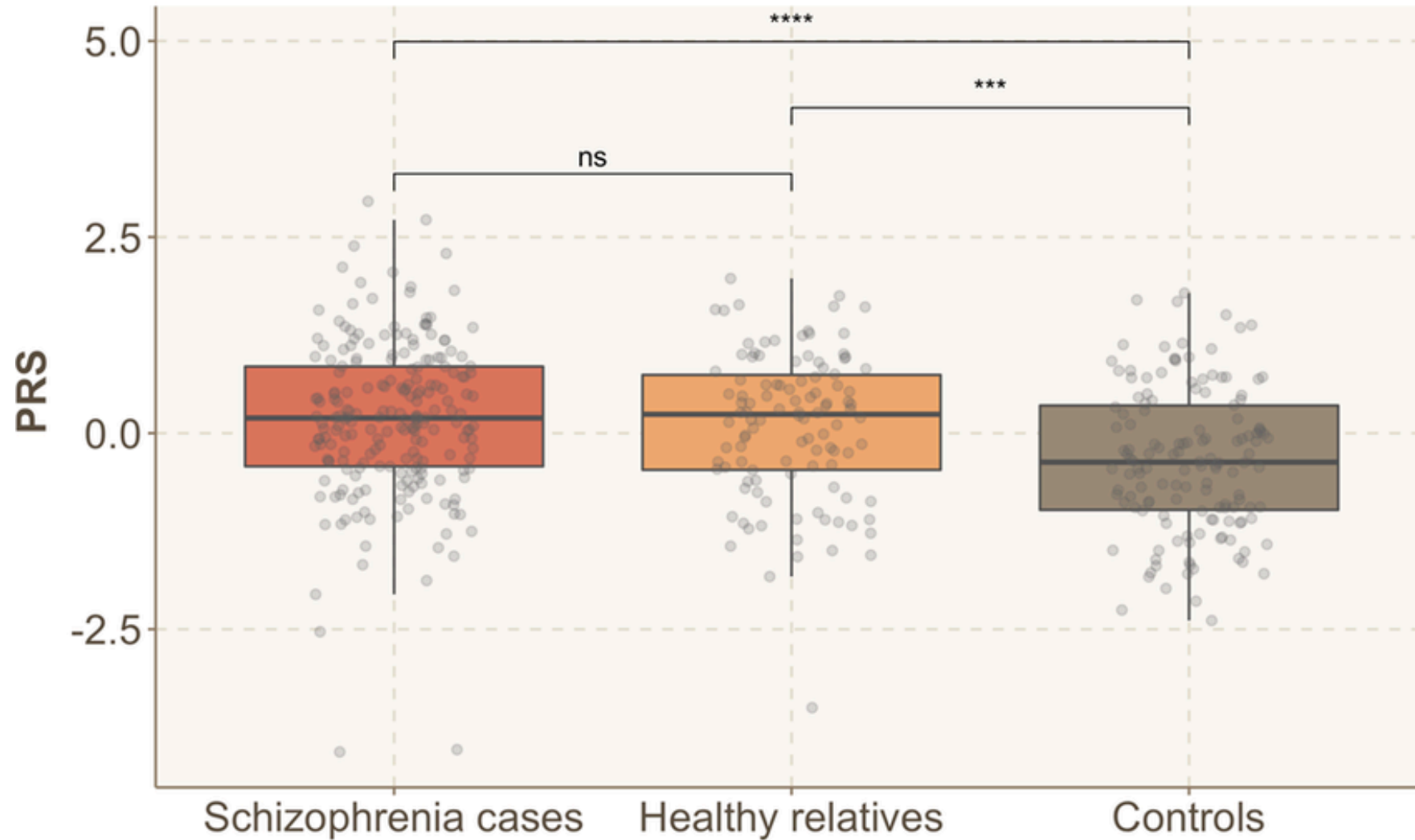
The estimate of β , $\hat{\beta}$, is called best linear unbiased estimator (BLUE) while the estimate of $\mathbf{g}$, $\hat{\mathbf{g}}$, is called best linear unbiased predictor (BLUP).

The polygenic score, $\hat{u}$, can be expressed as:

$$\hat{u} = \sum_{i=1}^M \mathbf{Z}_i \mathbf{g}_i$$

where, i represent SNP $(1, \dots, M)$

Polygenic risk score (PRS) for schizophrenia



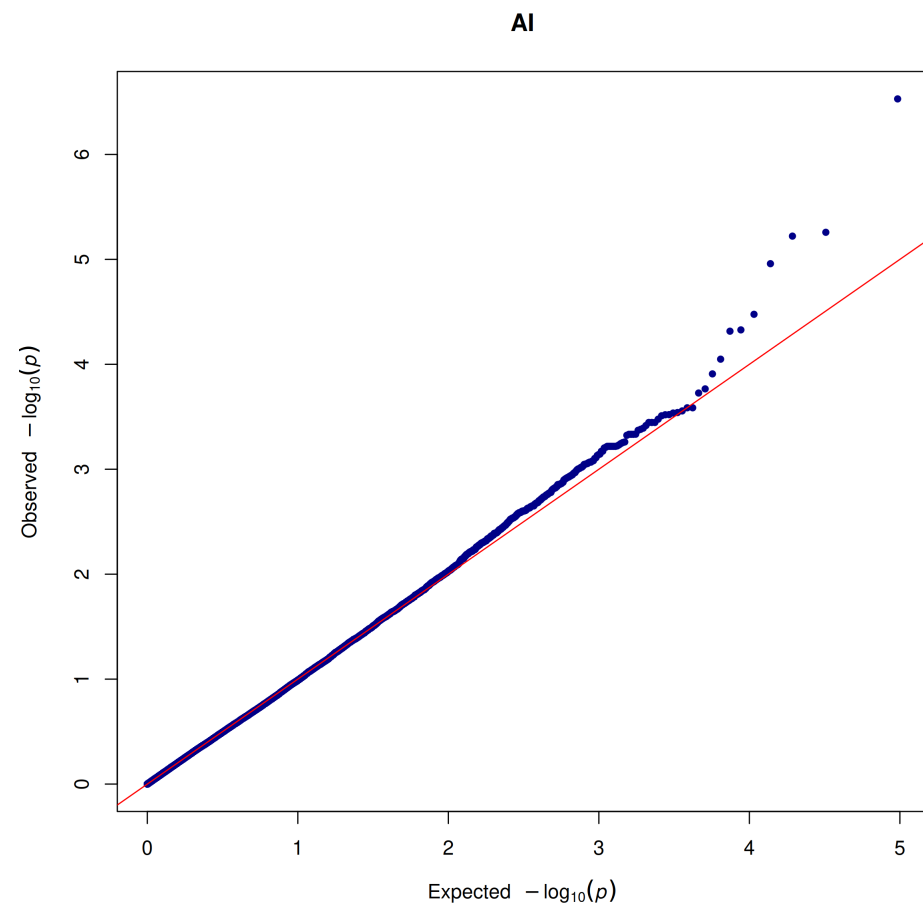
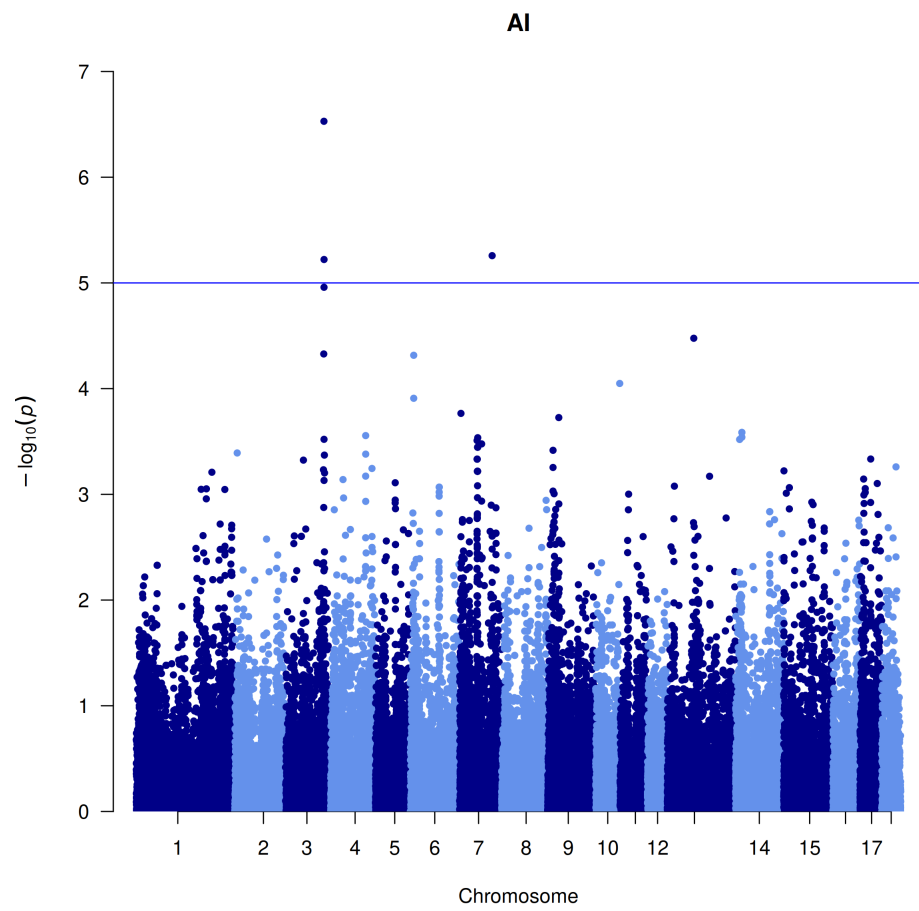
Shaqiri et al. *Translational Psychiatry* (2022)12:529

GWAS

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{S}\boldsymbol{\tau} + \mathbf{W}\mathbf{u} + \mathbf{e}$$

where, $\mathbf{y} \in \mathbb{R}^n$ is the phenotype of interest denoted as response variable; $\mathbf{X}$ is the $n \times p$ incidence/design matrix linking $\boldsymbol{\beta} \in \mathbb{R}^p$, the fixed effect(s) (e.g. sex) to the phenotype; $\mathbf{S}$ is the $n \times m$ incidence/design matrix linking $\boldsymbol{\tau} \in \mathbb{R}^m$, the fixed additive SNP effect, to the phenotype; $\mathbf{W} = \mathbf{I}$ (if the genotyped individuals have phenotypic information) is the $n \times n$ incidence/design matrix linking $\mathbf{u} \in \mathbb{R}^n$, the random polygenic effect, to the phenotype; and $\mathbf{e}$ is the residual variable.

The $\mathbf{u}$ and $\mathbf{e}$ vectors follows a normal distribution: $\mathbf{u} \sim \mathcal{N}(0, \sigma^2_u \mathbf{K})$, $\mathbf{e} \sim \mathcal{N}(0, \sigma^2_e \mathbf{I})$, and $\mathbf{K} = \frac{\mathbf{W}\mathbf{W}^\top}{m}$



Presentation Instruction

- Background: *Why and what did the authors do?*
- Materials and method: *How did they do it?*
- Results: *What did the authors found?*
- Conclusion: *What was the authors interpretation and implication of what they found?*

